6 **(a)** A student writes three incorrect statements as shown in the table below. Each statement has an error either in the unit or number. **Circle** the error in the statement and write the correct answer.

Incorrect statement	**Correct number or unit**
The weight of a person is about 700 kg.	
The atmospheric pressure at sea level is about 1.0×10^5 N m^2.	
1 GW is 10 times bigger than 1 MW.	

[3]

(b) For over a century, the standard kilogram has been defined by a small platinum-iridium cylinder housed at the International Bureau of Weights and Measures in France. This is the last remaining human-made material object on which a measurement standard is based.

Suggest why the use of a physical sample as the standard became inadequate with the advancement in Science.

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.....

..... [1]

(c) It was proposed at the turn of this century that mass be defined in terms of the Planck constant *h* rather than in terms of a physical standard mass. This can be done using a watt balance, an electromechanical mass measuring instrument that measures the mass of a test object very precisely by the strength of an electric current and a voltage.

Fig 6.1 shows a simplified version of the watt balance.

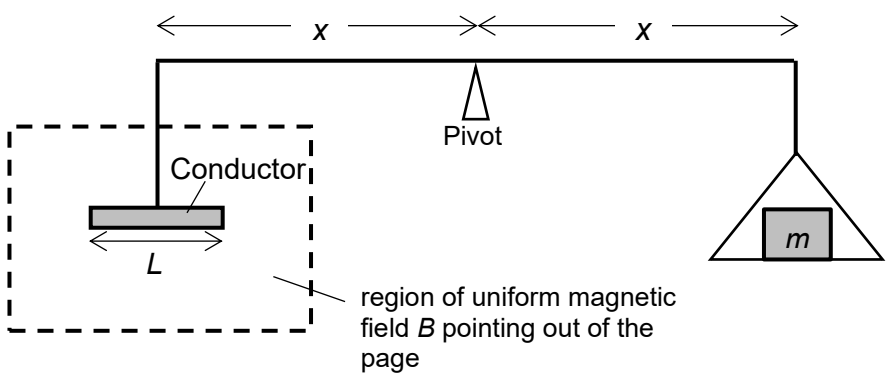

Fig 6.1

The mass *m* on the right arm of the balance is the mass to be measured. On the left arm, a conductor of length *L* is placed in a uniform magnetic field with flux density *B* pointing out of the page.

The process of finding *m* involves two stages.

(i) Stage 1: To balance the mass, a constant current I is passed through the conductor such that a magnetic force is exerted on the conductor. Using the principle of moments, the balancing condition is

$$mgx = BILx$$

where *g* is the gravitational field strength and can be treated as a constant.

Indicate the direction of the conventional current through the conductor by drawing an arrow on it in Fig. 6.1. [1]

(ii) Stage 2: The mass *m* is removed from the balance and the conductor is made to travel at a constant speed *u* through the same magnetic field as shown in Fig. 6.2.

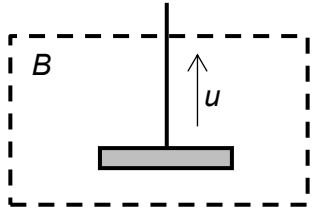

Fig. 6.2

1. Explain how a potential difference is set up aross the ends of the conductor.

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..... [2]

2. Hence, show that the potential difference *V* set up is given by *BLu*. [2]

(iii) Very precise and accurate values of V and I can be determined using $V = \frac{hC}{2e}$ and $I = \frac{kne}{2}$, where e is the elementary charge, n is an integer and C and k are known constants.

Using **(c)(i)** and **(ii)**, show how the mass m can be defined in terms of the Planck constant h. [2]